

Homework 4: VIDEO COMPRESSION

Image Compression Using 8x8 Block-Based DCT and Run-Length Encoding

1. Introduction

This report documents the implementation of a JPEG-like image compression method using an 8x8 block-based Discrete Cosine Transform (DCT) on the image “lena.png.” The compression includes steps such as quantization, zigzag scanning, run-length encoding, and decoding, followed by image reconstruction through inverse DCT. The final results are compared for two different quantization tables, assessing compression efficiency and image quality using PSNR (Peak Signal-to-Noise Ratio).

2. Problem Setup

The objective of this homework was to:

- Divide the grayscale “lena.png” image into 8x8 blocks and apply the DCT to transform spatial data into frequency components.
- Quantize the DCT coefficients using two distinct quantization tables to reduce data size by prioritizing perceptually important frequencies.
- Use a raster scan to process each 8x8 block in a consistent left-to-right, top-to-bottom order across the results of dividing process. For each block perform a zigzag scan and run-length encoding to compress each 8x8 block.
- Reconstruct the image by applying run-length decoding, inverse quantization, and inverse DCT to the compressed data.
- Compare the encoded sizes and PSNR of the reconstructed images for the two quantization tables.

3. Methodology

A. Image Division and 2D DCT

1. **Image Loading and Grayscale Conversion:** The image was converted to grayscale for processing simplicity.

Original Lena Image



Figure 1: Grayscale image loading step

2. **8x8 Block Division:** The grayscale image was divided into non-overlapping 8x8 blocks. Each block was then transformed into the frequency domain using the 2D DCT, which concentrates the energy of the block into a small number of low-frequency coefficients.

B. Quantization

Quantization tables were applied to compress the DCT coefficients by reducing the precision of high-frequency components:

Name: Le Minh Bang 黎明朋

StudentID: 313540015

10	7	6	10	14	24	31	37
7	7	8	11	16	35	36	33
8	8	10	14	24	34	41	34
8	10	13	17	31	52	48	37
11	13	22	34	41	65	62	46
14	21	33	38	49	62	68	55
29	38	47	52	62	73	72	61
43	55	57	59	67	60	62	59

Quantization table 1

10	11	14	28	59	59	59	59
11	13	16	40	59	59	59	59
14	16	34	59	59	59	59	59
28	40	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59
59	59	59	59	59	59	59	59

Quantization table 2

Figure 2: Quantization tables

- **Quantization Table 1:** Prioritized more detail retention with moderate compression.
- **Quantization Table 2:** Applied more aggressive compression, emphasizing lower frequencies and discarding more high-frequency data.

C. Zigzag Scanning and Run-Length Encoding

- **Zigzag Scanning:** Each 8x8 block's coefficients were rearranged in zigzag order to prioritize lower frequencies. This ordering facilitates run-length encoding by clustering zero-valued, high-frequency coefficients.

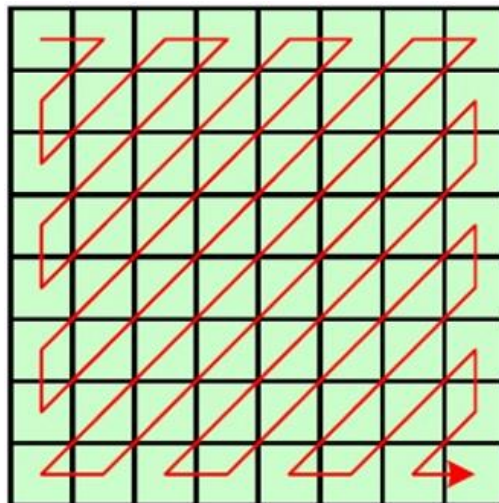


Figure 3: Zigzag scan visualization

- **Run-Length Encoding (RLE):** After the zigzag scan, RLE was applied to compress consecutive zeros, achieving significant data reduction.

D. Decoding and Reconstruction

1. **Run-Length Decoding and Inverse Zigzag Scan:** RLE was reversed to reconstruct the zigzag-scanned DCT coefficients.
2. **Inverse Quantization:** Each coefficient was multiplied by its respective value in the quantization table.
3. **Inverse DCT:** The 2D Inverse DCT was applied to each 8x8 block, converting the frequency data back into spatial pixel values.
4. **Recombination of Blocks:** All 8x8 blocks were recombined to produce the reconstructed image.

Reconstructed Image (Quantization Table 1)



Reconstructed Image (Quantization Table 2)



Figure 4: The result images

E. Compression Efficiency and Quality Assessment

- **Encoded Size Comparison:** The size of each compressed image was compared based on the total number of run-length encoded pairs.

- **PSNR Calculation:** PSNR values were computed to evaluate the visual quality of each reconstructed image.

4. Experiments

A. Reconstructed Images

The reconstructed images were generated for both quantization tables and displayed side-by-side with the original.



Figure 5: The result side by side of the original and reconstructed images

B. PSNR Analysis and Encoded Size Comparison

The PSNR values were calculated to assess the quality of the reconstructed images. Higher PSNR values indicate less compression-induced distortion.

Table 1: Comparison of two quantization tables:

Quantization Table	Encoded Size	PSNR (dB)
Table 1	146.07 KB	35.42 dB
Table 2	94.73 KB	32.23 dB

Quantization Table 1 preserved more detail, resulting in a higher PSNR, while Table 2 achieved more compression at the cost of reduced quality.

5. Conclusion

This homework successfully demonstrated the process of compressing an image using 8x8 block-based DCT, quantization, raster scan (blocks), zigzag scanning (pixels), and run-length encoding. The results reveal the trade-offs between compression and image quality:

- **Quantization Table 1** provided better image quality with a higher PSNR but resulted in a larger encoded size.
- **Quantization Table 2** achieved higher compression with a smaller encoded size but resulted in more noticeable quality loss.

In practical applications, the choice of quantization table and compression level would depend on the specific requirements, balancing storage needs with acceptable image quality. The methodology applied here provides a foundation for understanding JPEG-like compression techniques used in real-world image processing.